# Generative AI Application Development Using MLflow

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# Pre-requisite

**Familiarity with NLP terms**

Transformer

Prompt

LLM

Few shot learning

Eval

LLM-as-a Judge

Sessions require no prior MLflow familiarity.

# Session Flow

- Generative AI Basics
- GenAI App Development — traditional approach
- GenAI App Development — leveraging MLflow
- MLflow Features
- Demo
  - Notebook
  - Visual Studio project
  - Deployment and hosting

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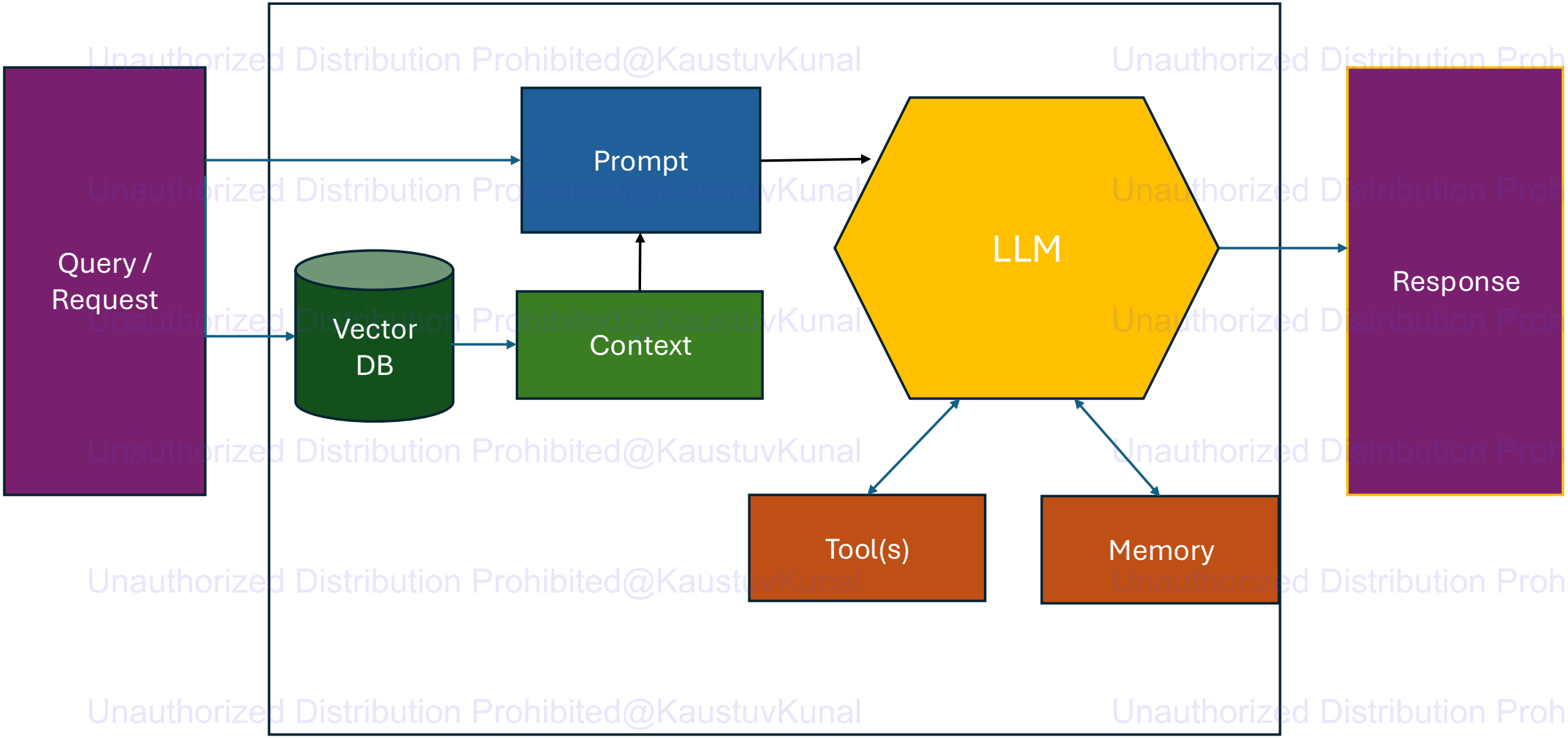
# Generative AI Applications (Vanilla →RAG→ Agentic)

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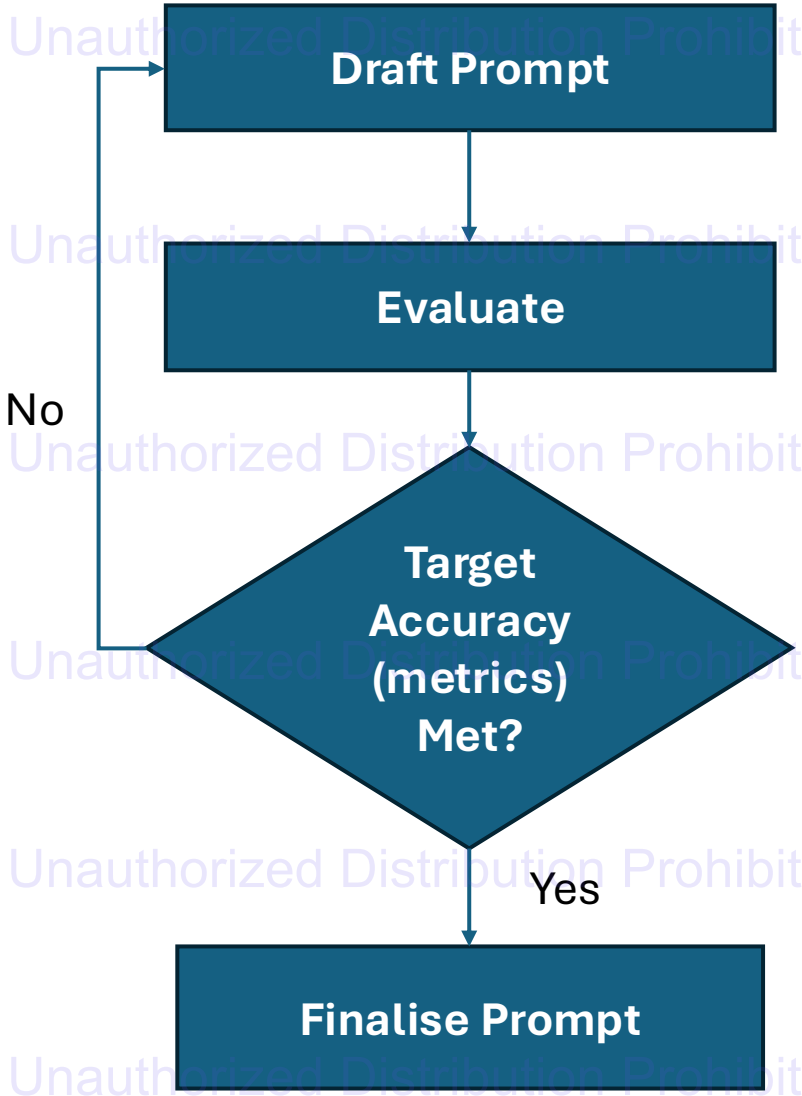
# GenAI Application Development: Traditional Approach




Eval

- Define Metric
- Eval Dataset

  Ground-Truth

  Golden-Records

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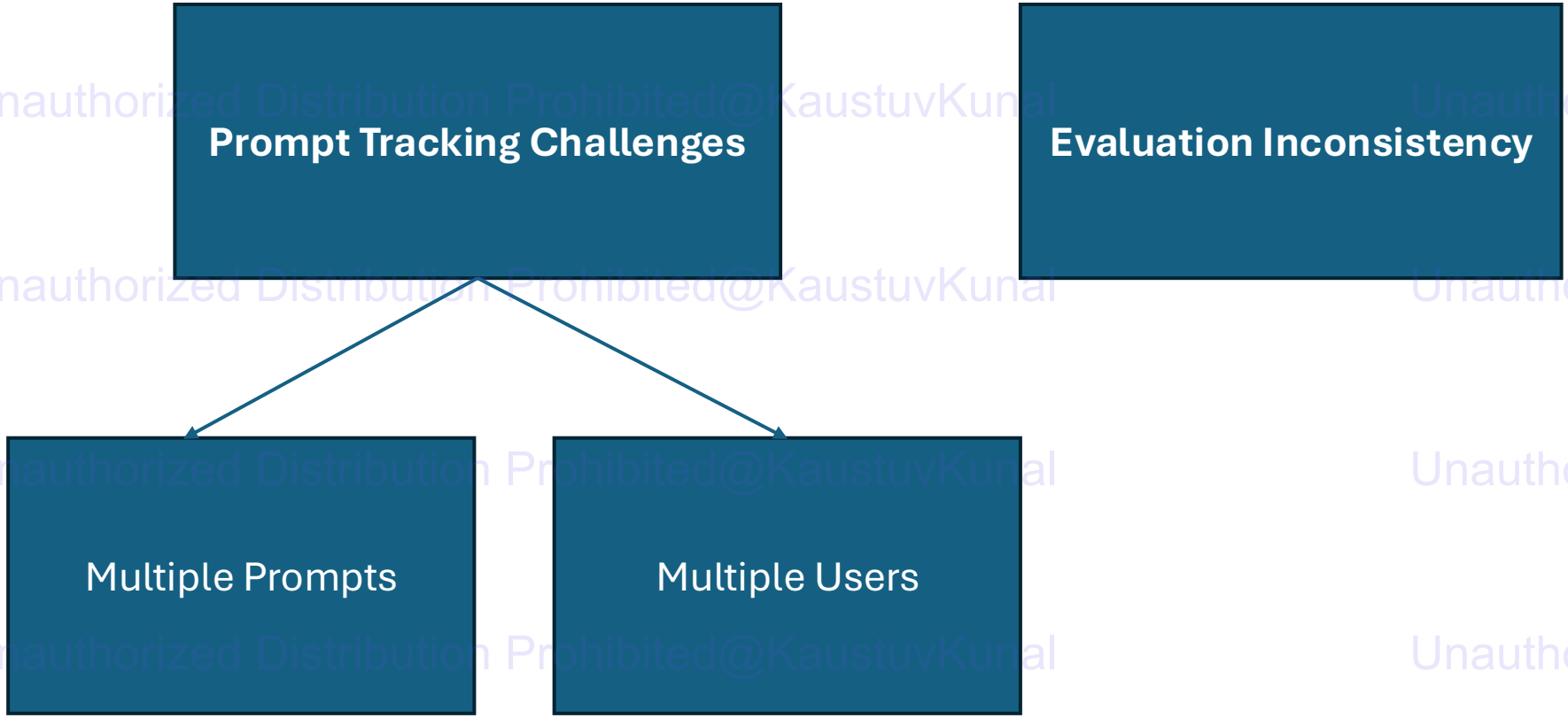
# Traditional Approach: Issues

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# Solution — Experiment Tracking Tool(s)

Weights & Biases

mlflow™

Langfuse

- Opensource
- Native GenAI Support
- Feature-rich

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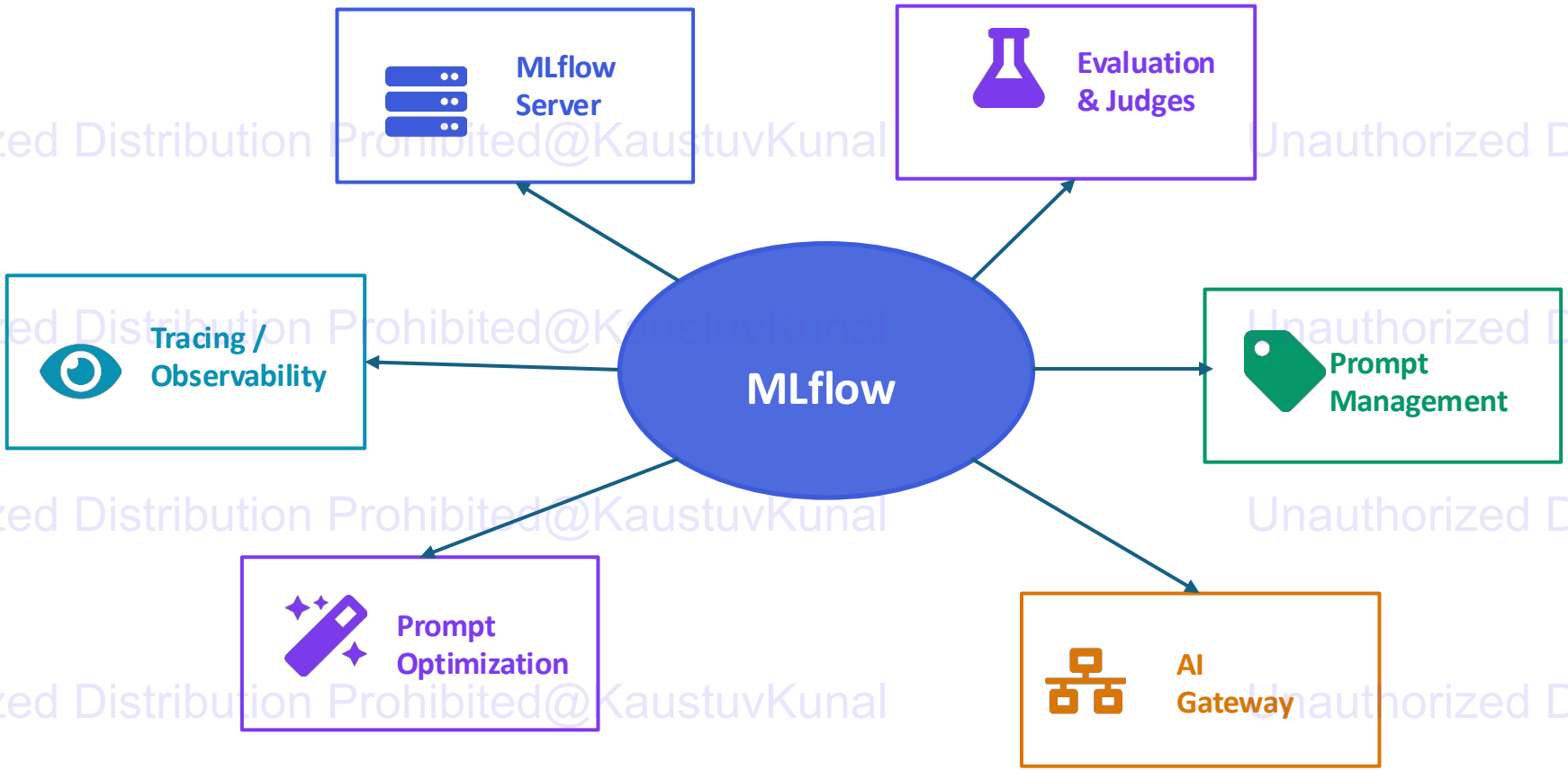
# MLflow — The AI Engineering Platform

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# Tracing /Observability

## Captures

- Step-by-step execution sequence
- Inputs
- Outputs
- Metadata

## Benefits

- Debugging
- Production Monitoring
- Evaluation
- Dataset Collection

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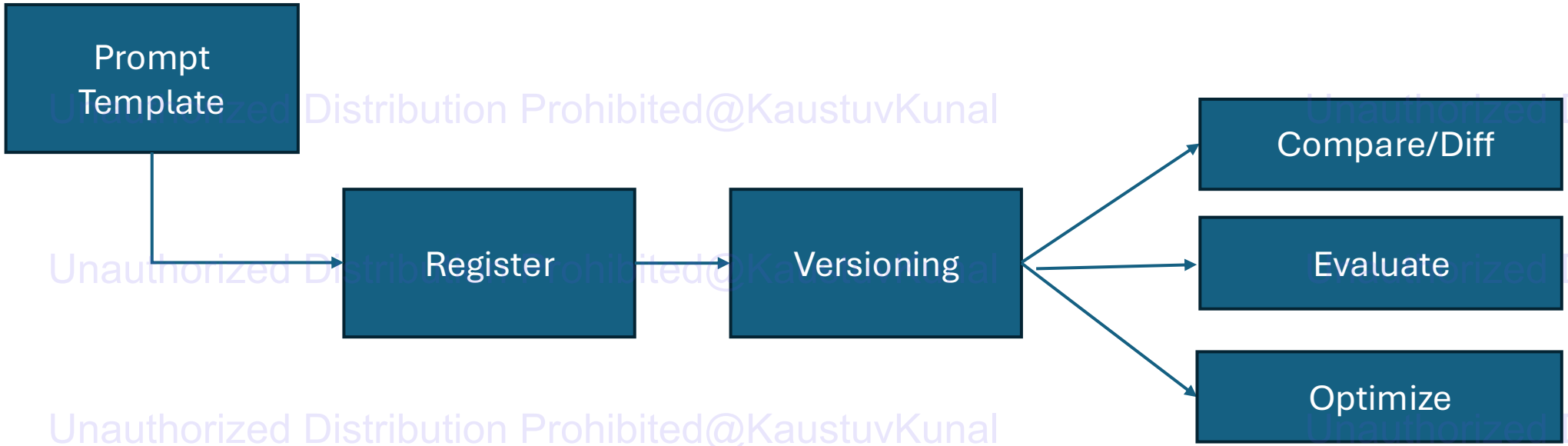
# Prompt Management

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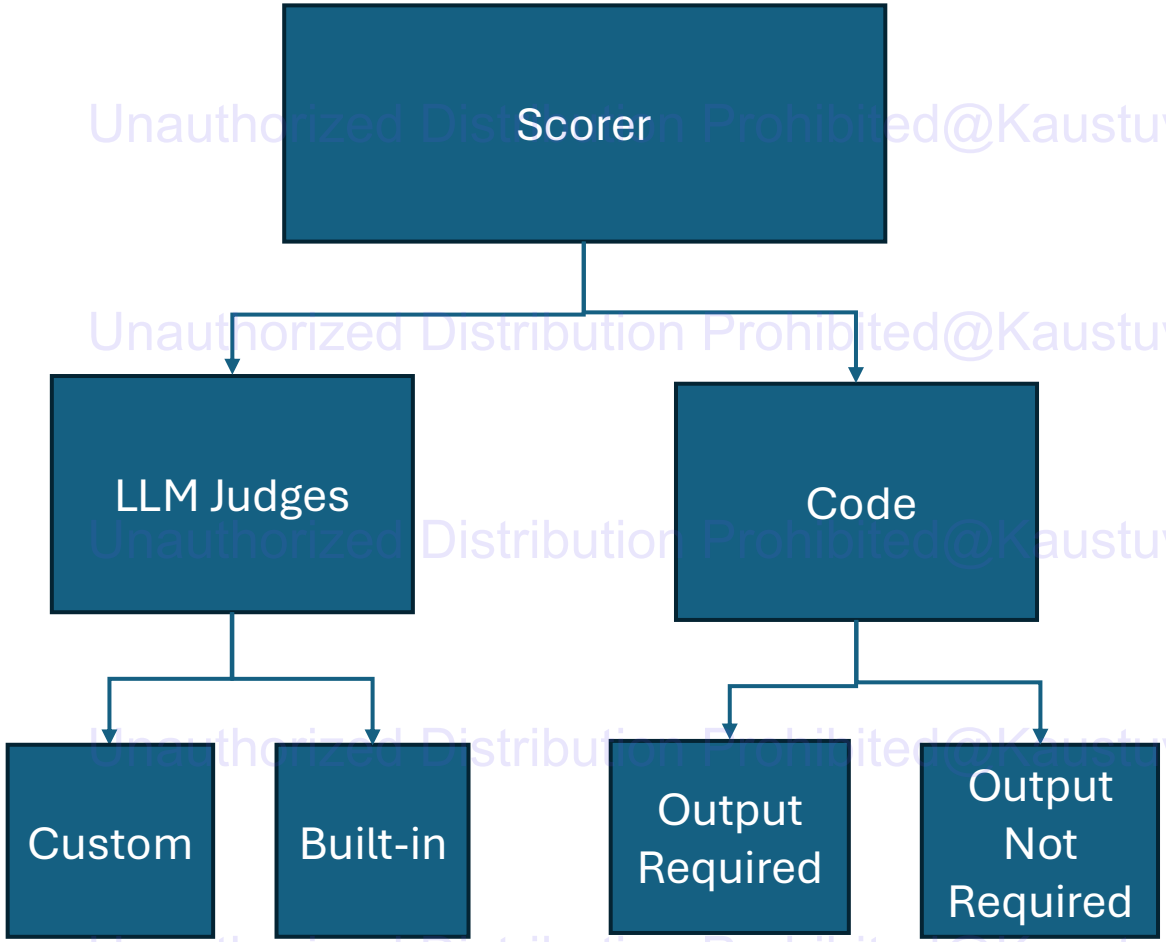
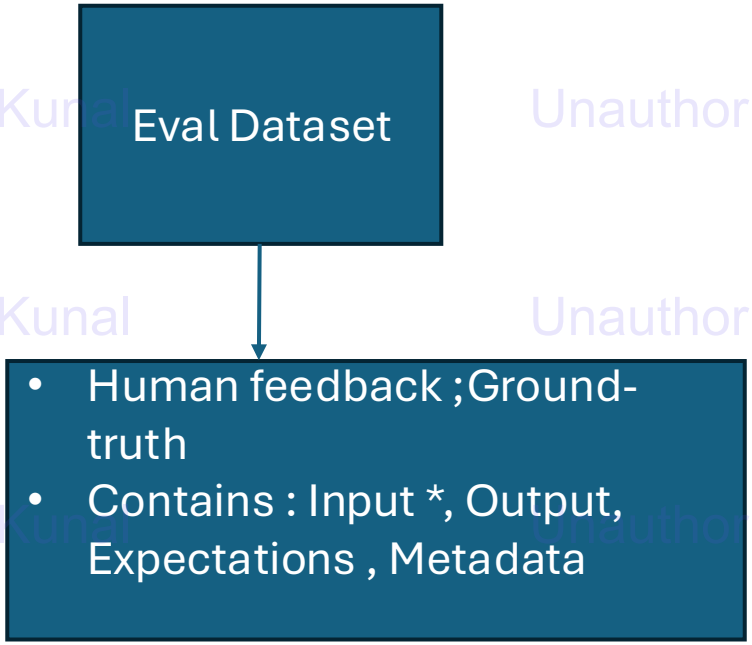
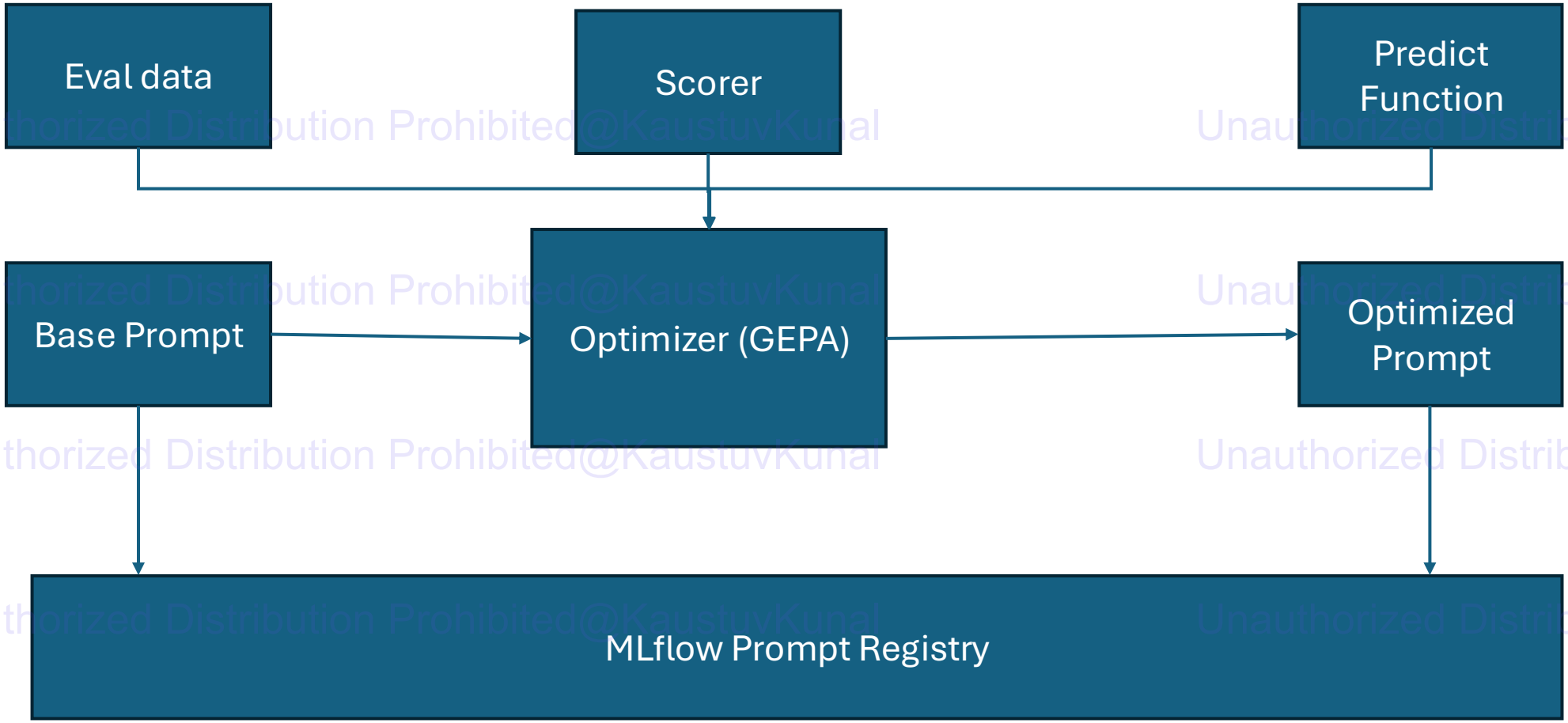
# Eval



# Prompt Optimization

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# Optimization Process

Evaluates current prompt on training examples

Analyse failure patterns

Generates improved prompt variations

Tests these variations to find best performer

Iterates until reach the maximum metric calls or convergence

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# Demo

https://github.com/kaustuvkunal/genai-support-ticket-classifier-mlflow-demo

# Optimization Disclaimers

- No guaranteed improvement -Prompt optimization does not always yield better results
- Eval data errors propagate - Flaws or biases in evaluation data directly affect the optimized prompt
- High compute cost & time -Requires significantly more LLM calls and longer processing time
- Inconsistent evaluation results- Scores may vary across runs due to LLM stochasticity
- Verbose optimized prompts -Often produces longer and more complex prompts

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# Q &A

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